Method

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Bins of size s samples, offset by o samples from the previous bin. o could be less than s.

Profile the data by computing a metric on the data in each bin. Naive approach uses max and min values of each metric across the population.

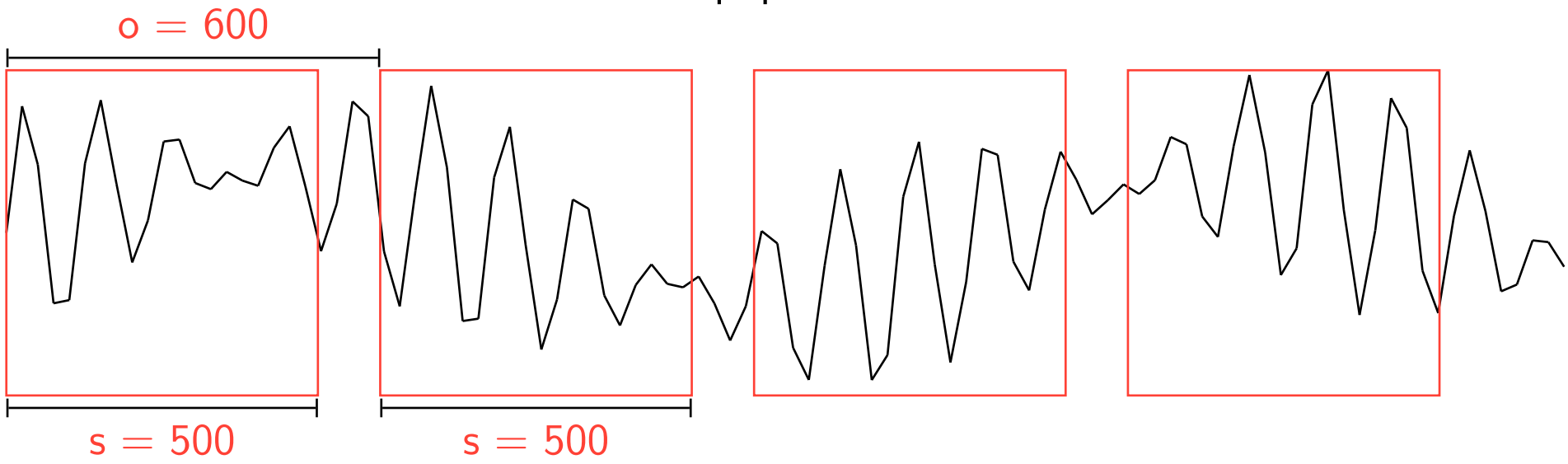


Figure 1: Bins with $o = 600$ and $s = 500$

The goal is to be able to uniquely distinguish each device. 2 metrics to evalute this:

absolute overlap:

$$\frac{\text{width of overlapping regions}}{\text{total width of values}}$$

distinguishability:

$$\frac{\text{number of distinguishable devices}}{\text{total number of devices}}$$

Method 1: by mean

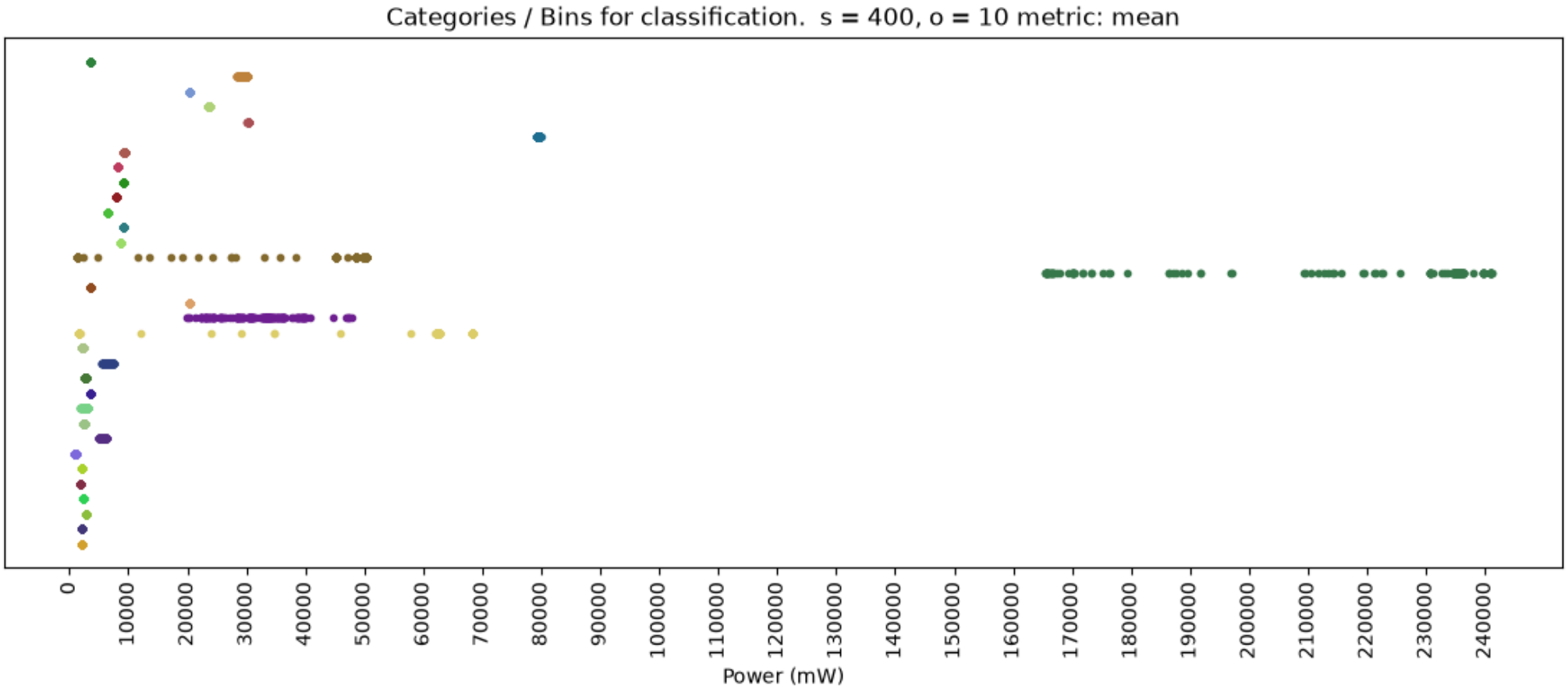
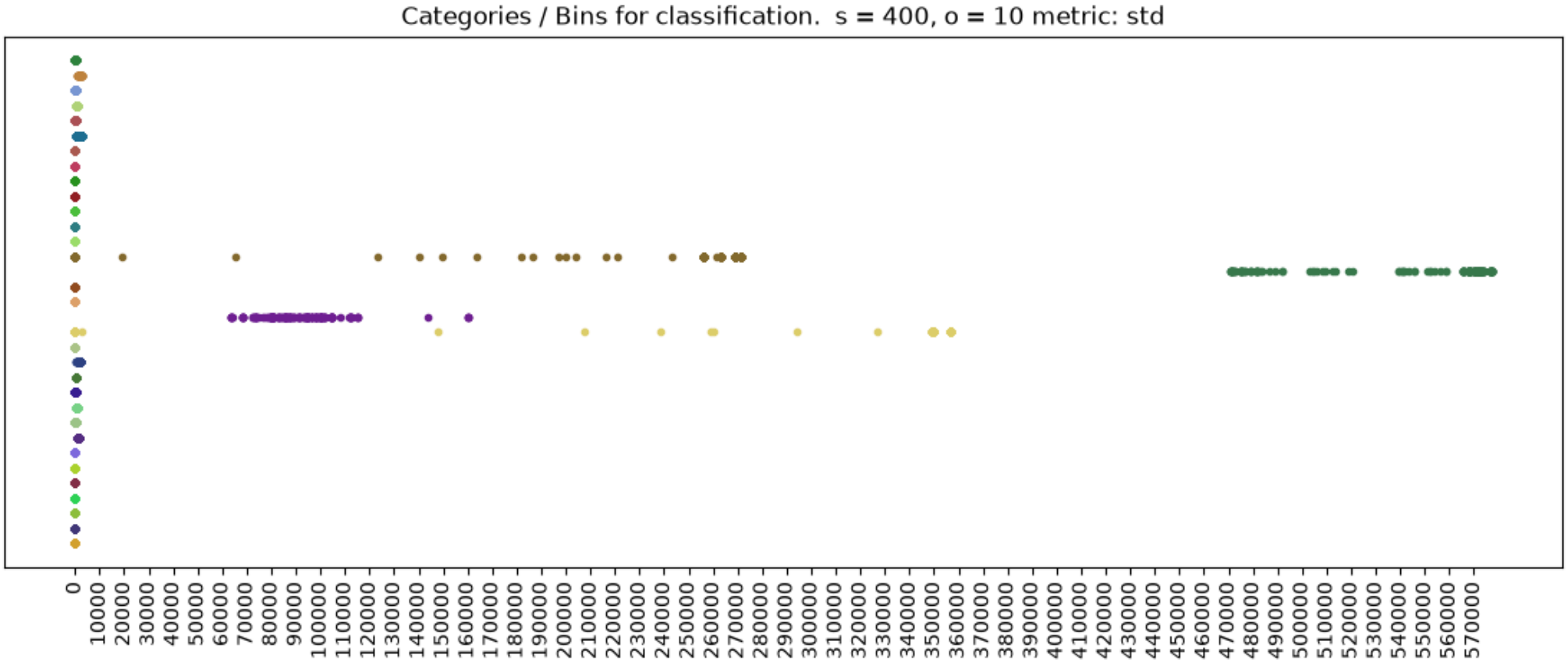


Figure 2: Y axis has no meaning, just to better show overlaps

Most devices are low powered, $<$ 50 Watts

Method 2: by std



Std overlaps more than mean, which makes sense since it doesn't carry information about the actual value, just it's spread

Method 3: Both

TODO: produce 2d graph of this

Evaluation:

	Method	absolute overlap	Distinguishable devices
0	Mean	x	3 / 33, (9.09%)
1	Std	x	1 / 33, (3.03%)
2	Both	x	x

Outlier Detection

apply this

Confusion Metric

table showing how it should be classified and what actually happened

Conformity Ttest

Test a range of values using colourmesh

Conclusions:

Mean can actually fully distinguish this dataset, with the bins set at $s = 500$ and $o = 10$.

Std cannot, which makes sense since it only measures the spread, no the actual value of result

Next Steps / Limitations

- Implement the absolute overlap metric
- Implement a metric using both
- Different values of o and s. Produce heatmap / colourmesh of this.
- Using only the min and max of a metric means that an outlier can throw off the performance. See if a filter or taking the meidan is better
- Explore fft?